

Linux Essentials - Week 2

Comprehensive Lecture Notes

Chapter 8: Finding Linux Documentation

Introduction

Definition: Linux provides built-in documentation to learn commands and troubleshoot issues.

Documentation Sources

Definition: Common sources are man, info, --help, whatis and apropos.

Important Commands:

- `man ls`
- `info ls`
- `ls --help`
- `whatis ls`
- `apropos copy`

Example:

Use 'man cp' to view the complete manual for the cp command.

Summary

Definition: Use man for detailed documentation and --help for quick syntax.

Knowledge Check

- What did you learn in this chapter?
- Practice the commands in a Linux terminal.

Summary

This chapter introduced the essential concepts and commands required for Linux administration.

Chapter 9: Processes

Introduction

Definition: A process is a program currently running in memory.

Process Attributes

Definition: Each process has a PID, owner, CPU usage and memory usage.

Important Commands:

- ps
- ps aux

Example:

List all running processes.

Monitoring

Definition: Monitor processes in real time.

Important Commands:

- top

Example:

Press q to quit.

Process Control

Definition: Terminate processes.

Important Commands:

- kill PID
- killall firefox

Example:

Kill a process by PID or name.

Knowledge Check

- What did you learn in this chapter?
- Practice the commands in a Linux terminal.

Summary

This chapter introduced the essential concepts and commands required for Linux administration.

Chapter 10: File Operations

Filesystems

Definition: A filesystem organizes files and directories.

Filesystem Layout

Definition: Important folders include /home, /etc, /bin, /var and /tmp.

Important Commands:

- `ls /`

Example:

View the root directory.

Comparing Files

Definition: Compare files using `diff` and `cmp`.

Important Commands:

- `diff a.txt b.txt`
- `cmp a.txt b.txt`
- `file a.txt`

Example:

Check differences and identify file type.

Backup & Compression

Definition: Archive with `tar` and compress with `gzip`.

Important Commands:

- `tar -cvf backup.tar folder`
- `tar -xvf backup.tar`
- `gzip file.txt`
- `gunzip file.txt.gz`

Example:

Create and extract backups.

Knowledge Check

- What did you learn in this chapter?
- Practice the commands in a Linux terminal.

Summary

This chapter introduced the essential concepts and commands required for Linux administration.

Chapter 11: Text Editors

Editors

Definition: Editors create and modify text files.

Nano & Gedit

Definition: Nano is terminal-based; Gedit is GUI-based.

Important Commands:

- nano file.txt
- gedit file.txt

Example:

Create or edit a text file.

Vi & Emacs

Definition: Vi/Vim and Emacs are advanced editors.

Important Commands:

- vi file.txt
- emacs file.txt

Example:

In vi: i=insert, Esc then :wq=save & quit.

Knowledge Check

- What did you learn in this chapter?
- Practice the commands in a Linux terminal.

Summary

This chapter introduced the essential concepts and commands required for Linux administration.

Chapter 12: User Environment

Users & Groups

Definition: Linux supports multiple users and groups.

Important Commands:

- whoami
- id
- groups

Example:

Display current user information.

Environment Variables

Definition: Variables such as PATH and HOME store system settings.

Important Commands:

- printenv
- echo \$PATH
- echo \$HOME

Example:

Display environment variables.

History

Definition: View previous commands.

Important Commands:

- history

Example:

Recall earlier commands.

Permissions

Definition: Read(r), Write(w), Execute(x).

Important Commands:

- ls -l
- chmod +x script.sh
- chown user file

Example:

Manage access permissions.

Knowledge Check

- What did you learn in this chapter?
- Practice the commands in a Linux terminal.

Summary

This chapter introduced the essential concepts and commands required for Linux administration.

Chapter 13: Manipulating Text

Text Utilities

Definition: Linux provides powerful text-processing commands.

Commands

Definition: View, search and edit text.

Important Commands:

- `cat file`
- `head file`
- `tail file`
- `grep Linux file`
- `sort file`
- `uniq file`
- `wc file`
- `cut -d: -f1 file`
- `tr a-z A-Z`

Example:

Search and manipulate text files.

sed & awk

Definition: Automate text editing and reporting.

Important Commands:

- `sed 's/Linux/Unix/' file`
- `awk '{print $1}' file`

Example:

Replace text and print columns.

Knowledge Check

- What did you learn in this chapter?
- Practice the commands in a Linux terminal.

Summary

This chapter introduced the essential concepts and commands required for Linux administration.

Chapter 14: Network Operations

Networking

Definition: Linux includes tools to configure and test networks.

Commands

Definition: View IP, test connectivity and transfer files.

Important Commands:

- `ip addr`
- `ping google.com`
- `nslookup google.com`
- `wget URL`
- `curl URL`
- `ssh user@host`
- `scp file user@host:/home`

Example:

Connect to remote systems securely.

Knowledge Check

- What did you learn in this chapter?
- Practice the commands in a Linux terminal.

Summary

This chapter introduced the essential concepts and commands required for Linux administration.

Chapter 15: Bash Shell and Basic Scripting

Bash

Definition: Bash is the default Linux shell used to execute commands and scripts.

Basic Script

Definition: A shell script automates repetitive tasks.

Important Commands:

- `#!/bin/bash`
- `echo Hello World`

Example:

Save as `script.sh`, then run `chmod +x script.sh` and `./script.sh`

Variables & Input

Definition: Store values and accept user input.

Important Commands:

- `name='Akshay'`
- `echo $name`
- `read name`

Example:

Print entered value.

Knowledge Check

- What did you learn in this chapter?
- Practice the commands in a Linux terminal.

Summary

This chapter introduced the essential concepts and commands required for Linux administration.

Chapter 16: More Bash Scripting

String Manipulation

Definition: Modify and display strings.

Important Commands:

- `echo ${name}`

Example:

Print variable.

Case & Loops

Definition: Decision making and repetition.

Important Commands:

- `case $1 in start) echo Start;; esac`
- `for i in 1 2 3; do echo $i; done`
- `while true; do break; done`

Example:

Examples of control structures.

Debugging

Definition: Find script errors.

Important Commands:

- `bash -x script.sh`

Example:

Run script in debug mode.

Knowledge Check

- What did you learn in this chapter?
- Practice the commands in a Linux terminal.

Summary

This chapter introduced the essential concepts and commands required for Linux administration.

Chapter 17: Printing

Printing

Definition: Linux supports command-line printing.

Commands

Definition: Print and manage print jobs.

Important Commands:

- `lp file.pdf`
- `lpr file.pdf`
- `lpq`
- `cancel JOBID`
- `ps2pdf file.ps`

Example:

Submit and manage print jobs.

Knowledge Check

- What did you learn in this chapter?
- Practice the commands in a Linux terminal.

Summary

This chapter introduced the essential concepts and commands required for Linux administration.

Chapter 18: Local Security Principles

Security

Definition: Protect Linux systems using permissions and least privilege.

Commands

Definition: Manage passwords and administrator access.

Important Commands:

- `passwd`
- `sudo` command
- `sudo -i`
- `chmod 755 file`
- `chown user file`

Example:

Secure users and files.

Best Practices

Definition: Use strong passwords, update software, avoid unnecessary root access.

Knowledge Check

- What did you learn in this chapter?
- Practice the commands in a Linux terminal.

Summary

This chapter introduced the essential concepts and commands required for Linux administration.